Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	MindScope: An Autonomous Agent for Deep Research and Business Reporting			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), **AI (√)**, Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Software Solutions	Management Strategy / Finance / Trade Services	Knowledge Creation / Distribution / Evaluation Technologies
Concept of Proposed Technology	**Context.** Korean SMEs currently face a "data utilization gap." They possess massive amounts of unorganized documents. However, they lack the tools to extract deep insights from them. Current AI solutions cannot reliably answer complex, multi-hop questions. They also struggle to generate comprehensive reports with accurate citations. **Core Idea.** To solve this, we propose "MindScope." Existing tools rely on linear retrieval, which often leads to logical dead ends when answering complex questions. MindScope overcomes this by modeling research as a **dynamic tree exploration**. Unlike linear search, this structure allows the agent to explore multiple hypotheses simultaneously and **backtrack** if a path proves irrelevant, ensuring deep logical coverage. This process is optimized by an **LLM guidance policy**, which acts as a "look-ahead" critic to predict the value of data before reading it. By strictly prioritizing "Top-of-Mind" information relevant to the current subtask, the system enables deep, multi-hop reasoning without overflowing the context window.			

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<td rowspan="3">Rationale for Joint R&D Project</td>
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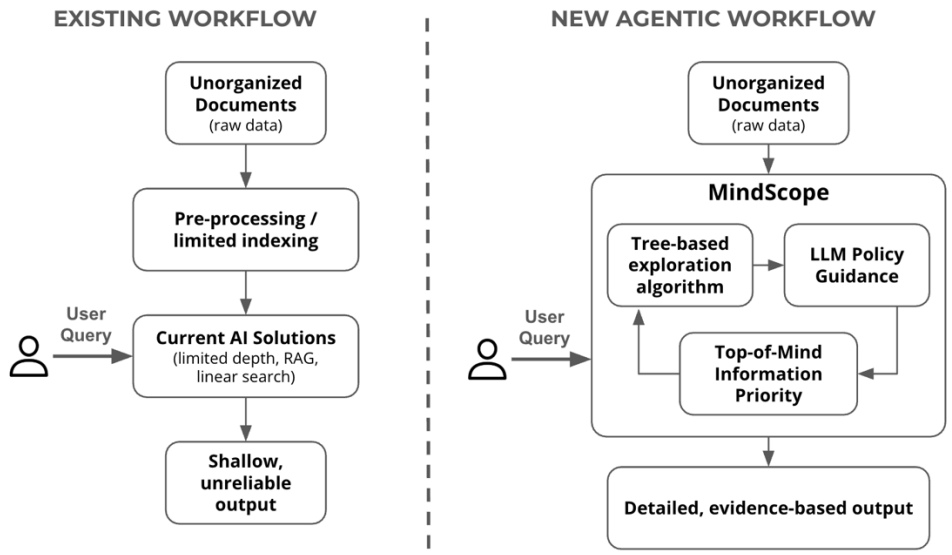

Figure 1. Current solutions struggle with depth and accuracy due to rigid retrieval paths (Left), MindScope (Right) integrates tree-based exploration with LLM-guided planning; it autonomously identifies, verifies, and synthesizes complex information into actionable business insights.

Expected Impact. The result is an instant "AI Analyst" capability for SME vendors. They can deploy solutions that generate detailed, evidence-based reports without expensive setup. This democratizes high-level business intelligence for sectors like legal and finance.
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Overall Benefit. This joint R&D initiative bridges the critical gap between theoretical innovation and commercial viability. It accelerates the development of the "MindScope," transforming advanced agentic research into a practical, "Zero-Setup" solution that democratizes high-level business intelligence for the Korean market.

How University Research Can Help. SMEs Small and medium-sized enterprises often lack the R&D resources to experiment with high-risk, cutting-edge technologies like Tree-Based Exploration and LLM Guidance Policies. The university provides this foundational technology, offering SMEs a sophisticated reasoning engine that overcomes the depth and reasoning limitations of current off-the-shelf tools without the prohibitive development costs.

How SMEs Can Facilitate University Research. Academic models frequently struggle with the complexities of real-world data. The SME partner is essential for providing domain-specific datasets (e.g., actual legal or financial documents) and commercial testing environments. This allows the university to validate and refine its algorithms against messy, real-world scenarios, ensuring the "Top-of-Mind" scoring and subtask planning mechanisms are robust enough for actual business use.
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<td>Keywords</td>
<td>
1. AI Agent
2. Information Retrieval
3. Database Systems
4. Finance, law, accounting, medicine
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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (7)

R&D Objectives

Current Market Opportunity & Pain Points

Korean SME software vendors are racing to integrate AI into their B2B products (e.g., LegalTech, MedTech, EduTech). However, they currently rely on simple "wrappers" around OpenAI or Claude APIs. This creates two critical risks:

- ***Lack of Differentiation*:** Everyone offers the same generic "Chat with PDF" features. The barrier to entry is too low.
- ***Inability to Handle Complex Workflows*:** End-clients (e.g., law firms, hospitals) do not want simple chatbots; they need "Agents" that can draft compliance reports, audit supply chains, or synthesize research. Current vector-based retrieval (RAG) fails at these multi-step reasoning tasks because it cannot connect distant dots without massive hallucination.
- ***The "Setup" Barrier:*** To get better results, vendors currently have to build complex Knowledge Graphs or fine-tune models, which is too expensive and slow for agile B2B sales cycles.

Core Innovation (The "Dynamic Navigator")

To solve these high-level market problems, we propose a **State-Aware Autonomous Agent** that removes the need for pre-built indexes or static graphs. Instead of searching a database, the agent treats the raw data (thousands of PDFs) as an environment to be explored. When a query is received, the agent dynamically navigates through the documents, building a "**Reasoning Trace**" in real-time. This allows SME vendors to offer "Zero-Setup" intelligence that is far deeper than standard RAG.

Specific Objectives:

1. ***Develop the "Cognitive State Representation" (Fundamental Tech).***
   a. Create a novel mathematical representation of the agent's "mind" during a search. This state must track: (i) Accumulated Knowledge (What facts I have gathered), (ii) Search Horizon (Which document links are promising), (iii) Logical Gaps (What is missing to answer the user's specific question).
   b. This enables the agent to stop when confident or backtrack when stuck, mimicking human research behavior.
2. ***"Zero-Setup" Architecture for B2B Scalability:***
   a. Eliminate the need for complex ETL (Extract, Transform, Load) pipelines. The objective is **"Ingest & Solve."**
   b. SME vendors can pitch "Instant Analysis" products where end-clients dump raw files and get immediate deep insights.
3. ***Traceability via Dynamic Graph Construction:***
   a. While the agent explores, it constructs a temporary "**Reasoning Trace Graph**." This is the deliverable. It maps exactly which documents were visited and how they contributed to the answer, providing 100% explainability for B2B compliance.

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R&D Target Specifications	Our primary goal is to build a "Dynamic Cognitive State Engine" that overcomes the limitations of current pre-indexed/static solutions. We have defined six key quantitative targets that will ensure this technology provides a tangible competitive advantage to the SME vendors who adopt it: **1. Setup Latency (Time to First Query)** • **Target:** < 5 Seconds (Zero-Index) • **Significance:** Current solutions often require hours of indexing or graph building before a user can ask a question. By achieving near-instant readiness, we enable our partners to sell "Instant Deployment" solutions where clients can upload data and immediately begin analysis. **2. Reasoning Depth (Logical Hops)** • **Target:** 10+ Steps • **Significance:** Standard vector retrieval is typically effective for only ~3 logical hops. By enabling the agent to follow reasoning trails as deep as necessary—similar to a human auditor—we solve the "Complex Workflow" pain point that prevents current bots from handling deep analysis. **3. State Awareness (Uncertainty Detection)** • **Target:** Self-Confidence Score Output • **Significance:** Current models often hallucinate answers when data is missing. Our agent will calculate and output a confidence score based on its internal state stability. This allows the system to explicitly flag "I don't know" or "Data is insufficient," which is critical for preventing liability in LegalTech and MedTech applications. **4. Context Management (Information Capacity)** • **Target:** Recursive State Update (Effectively Infinite) • **Significance:** Instead of a fixed context window that limits reading to small batches of files, our agent compresses its reading history into a continuously updating state vector. This allows it to process an unlimited number of documents sequentially without losing the "thread" of the investigation. **5. Explainability (Audit Trail)** • **Target:** Visual Decision Tree Generation • **Significance:** B2B clients require proof, not just answers. Unlike black-box LLMs that list generic references, our engine will output the exact traversal path (e.g., "Visited Document A → Found Link → Visited Document B"), providing 100% explainability for compliance purposes. **6. Cost Efficiency (Retrieval Precision)** • **Target:** High Precision via Selective Reading • **Significance:** Standard RAG retrieves and processes many irrelevant chunks, driving up costs. Our agent uses a policy network to "predict" which document is worth reading next, significantly reducing token consumption. We target a >40% reduction in API costs compared to standard exhaustive search methods.
R&D Plan	*Timeline: 24 Months* **Phase 1: The Agentic Tree Exploration Framework (Months 1–8)** *Focus: Building the fundamental engine where the agent explores data as a growing tree structure.* **Task 1.1: Tree-Based Exploration Environment** • **Description:** We will develop the core backend where the agent's search history is modeled as a dynamic **Exploration Tree**. The root is the initial user query, and each branch represents a "navigation action" (e.g., Open Document A, Search Keyword X, Read Paragraph 3). • **Core Innovation:** Most current agents use linear chains (ReAct, Yao et al., 2023). We adopt the tree-search methodology of LATS (Language Agent Tree Search) (Zhou et al., 2023) but

innovate by scaling it from simple reasoning puzzles to **massive external document retrieval**. Unlike GraphReader (Li et al., 2024) which explores a static graph, our tree is constructed *dynamically* based on the agent's unique navigation path.

Task 1.2: LLM-Based Guidance Function

- **Description:** Implement a Guidance Policy that uses an LLM to evaluate the leaf nodes of the tree and determine which node is most promising to expand next.
- **Core Innovation:** We integrate the "Self-Reflection" mechanism from Self-RAG (Asai et al., 2024). However, instead of just critiquing the final answer, our Guidance Function acts as a "Look-Ahead" critic, estimating the information gain of a document before paying the cost to fully read it, significantly optimizing token usage compared to standard RAP (Reasoning via Planning) (Hao et al., 2023).

Task 1.3: Incremental Report Generation

- **Description:** Develop a "writer" module that synthesizes information found at visited nodes into a coherent report, updating the draft in real-time.
- **Core Innovation:** This moves beyond "Retrieved-Then-Generate" pipelines. By generating incrementally, we allow the agent to detect "Information Saturation" early—a concept aligned with Adaptive-RAG (Jeong et al., 2024)—stopping the search the moment the report is sufficient.

Phase 2: "Top-of-Mind" Context Optimization (Months 9–16)

Focus: Solving the context window bottleneck via dynamic scoring and subtask awareness.

Task 2.1: Subtask Planning & Decomposition

- **Description:** Implement a high-level Chain-of-Thought (CoT) Planner that breaks the user goal into sequential Subtasks (e.g., Goal: “Analyze Risk" Subtask 1: "Find Financial Liabilities" Subtask 2: "Check Legal Compliance").
- **Core Innovation:** We extend the Plan-and-Solve framework (Wang et al., 2023). The innovation is the persistence of the Active Subtask State. Unlike HippoRAG (2024), which uses static Personalized PageRank for retrieval, our agent dynamically alters its retrieval criteria based entirely on which is currently active.

Task 2.2: Subtask-Aware Node Scoring

- **Description:** Develop a scoring function that evaluates every node in the tree against the current subtask.
- **Core Innovation:** This utilizes the "Context Pruning" principles from Fewer is More (Huang et al., 2024). We innovate by making the pruning conditional. A node containing Revenue Data may be pruned (low score) during the Legal Check subtask but is immediately resurrected (high score) when the agent switches to the Financial Analysis subtask. This Resurrection capability is absent in standard sliding-window memory models like MemWalker (Chen et al., 2023).

Task 2.3: "Top-of-Mind" Context Construction

- **Description:** To address context limits, we implement a "Top-of-Mind" Selector. Before every decision step, the system ranks all nodes using the scoring function from Task 2.2. Only the top-nodes plus the Active Path are serialized and fed to the LLM.
- **Core Innovation:** This creates a "Working Memory" analogous to the Generative Agents memory stream (Park et al., 2023), but optimized for high-fidelity document segments rather

	than social simulation summaries. It ensures the LLM is always "thinking about" the specific data points most relevant to the immediate subtask. **Phase 3: Full-Stack Integration & Multi-Domain Validation (Months 17–24)** *Focus: Visualization and rigorous testing across diverse verticals.* **Task 3.1: "Glass Box" Frontend Development** • **Description:** Develop a user interface focused on Explainability (XAI). The UI will visualize the Exploration Tree, the Current Subtask, and the "Top-of-Mind" nodes. • **Core Innovation:** Most tools like ChatGPT are "Black Boxes." We provide a "Reasoning Trace Visualization" that allows B2B users to audit exactly *why* the agent chose to expand Node B over Node C, a requirement for high-stakes domains (legal/medical). **Task 3.2: Multi-Domain Evaluation** • **Description:** Benchmark the system on Legal (Precedent Search), Medical (Diagnostic Guidelines), and Finance (Risk Analysis). • **Metric:** We will compare against GraphReader and Self-RAG, targeting a 20% higher accuracy in finding the correct answers. **Task 3.3: Final System Optimization** • Description: Optimize latency of the Guidance function to ensure a responsive "co-pilot" experience.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute (UIUC) will lead the **core algorithmic development** and **theoretical framework** of the MindScope engine. • **Core Engine Development:** Design and implement the Tree-Based Exploration Algorithm that enables non-linear navigation of raw data. • **Guidance Policy Design:** Develop the LLM Guidance Policy and the "Top-of-Mind" Scoring Function to dynamically prioritize nodes based on active subtasks. • **Context Optimization:** Create the mathematical model for Context Pruning to ensure high-fidelity reasoning within token limits. • **Academic Benchmarking:** Conduct rigorous evaluation against state-of-the-art baselines (e.g., GraphReader, Self-RAG) using standard datasets to validate improvements in depth and reasoning accuracy.
Expected R&R of Korean SME Partner	The Korean SME Partner will lead the **system integration**, **commercialization**, and **real-world validation** of the technology. • **Domain Data Provision:** Secure, anonymize, and curate high-stakes B2B datasets (legal precedents, financial audit logs, medical guidelines) to train and test the "Critic" models in real-world scenarios. • **"Glass Box" Frontend Development:** Build the commercial user interface that visualizes the Reasoning Trace and "Top-of-Mind" highlights, ensuring auditability for B2B users. • **Full-Stack Integration:** Integrate the core research code into a scalable, secure commercial backend suitable for enterprise deployment (SaaS architecture). • **Pilot Validation:** execute Pilot Programs with actual industry clients to measure "Time-to-Insight" and "Subtask Completion Rate," providing feedback for algorithm refinement.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** • **MindScope Engine Validation:** Successful development of the first commercial-grade agentic engine capable of **non-linear tree exploration** on raw, unorganized data, surpassing the limitations of linear RAG systems. • **"Top-of-Mind" Context Efficiency:** Proof that dynamic subtask-aware node pruning can reduce token usage by ~20% while significantly improving accuracy on complex, multi-hop queries. • **Explainable AI Standard:** Establishment of the "**Glass Box**" Reasoning Trace as a new industry standard for auditability, allowing users to verify every logical step and citation. **Commercialization Potential** • **"Zero-Setup" B2B Solution:** Creation of a plug-and-play API that enables SME vendors in LegalTech, FinTech, and Medical sectors to deploy "AI Analyst" capabilities instantly, without the high cost of building Knowledge Graphs. • **New Revenue Streams:** Enabling Korean SME vendors to transition from simple "search providers" to "intelligence partners," allowing them to charge for high-value automated reporting and risk analysis. **Industrial and Economic Impact** • **Bridging the Data Gap:** Empowering Korean SMEs to compete with large enterprises by unlocking the massive value currently trapped in their dormant, unorganized document repositories. • **Productivity Revolution:** Reducing the time professionals spend on manual document research by an estimated 60-70%, freeing human talent for strategic decision-making. • **Trust in AI Adoption:** Overcoming the "Black Box" barrier in regulated industries by providing fully traceable, evidence-based AI outputs, thereby accelerating safe AI adoption across the Korean economy.

6. References

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(8) Wang, L., et al. (2023). Plan-and-Solve Prompting: Improving Zero-Shot Chain-of-Thought Reasoning by Large Language Models. ACL 2023.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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